Schritt 1 – Apache Webserver installieren

```
student@debian: ~
student@debian:~$ sudo apt install apache2
[sudo] Passwort für student:
Paketlisten werden gelesen… Fertig
Abhängigkeitsbaum wird aufgebaut… Fertig
Statusinformationen werden eingelesen… Fertig
apache2 ist schon die neueste Version (2.4.62-1~deb11u2).
0 aktualisiert, 0 neu installiert, 0 zu entfernen und 88 nicht aktualisiert.
student@debian:~$
```

```
student@debian:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
student@debian:~$ sudo systemctl restart apache2
student@debian:~$
```

Schritt 2 – HTTPS aktivieren (Port 443)

Schritt 3 – Selbstsigniertes Zertifikat erstellen

```
student@debian:~$ sudo mkdir /etc/apache2/ssl
student@debian:~$ sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout /etc/apache2/ssl/naw5.key \
  -out /etc/apache2/ssl/naw5.crt
Generating a RSA private key
................+++++
.............+++++
writing new private key to '/etc/apache2/ssl/naw5.key'
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:DE
State or Province Name (full name) [Some-State]:furtwangen
Locality Name (eg, city) []:.
Organization Name (eg, company) [Internet Widgits Pty Ltd]:.
Organizational Unit Name (eg, section) []:.
Common Name (e.g. server FQDN or YOUR name) []:srever5
Email Address []:.
student@debian:~$
```

Habe einen fehler name soll **www.naw5.informatik.hs-furtwangen.de**

Schritt 4 – HTTPS VirtualHost konfigurieren

```
  GNU nano 5.4                                  /etc/apache2/sites-avai
<IfModule mod_ssl.c>
    <VirtualHost *:443>
        ServerAdmin webmaster@naw5.informatik.hs-furtwangen.de
        ServerName www.naw5.informatik.hs-furtwangen.de
        DocumentRoot /var/www/html

        SSLEngine on
        SSLCertificateFile /etc/apache2/ssl/naw5.crt
        SSLCertificateKeyFile /etc/apache2/ssl/naw5.key

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
    </VirtualHost>

    <VirtualHost [::]:443>
        ServerAdmin webmaster@naw5.informatik.hs-furtwangen.de
        ServerName www.naw5.informatik.hs-furtwangen.de
        DocumentRoot /var/www/html

        SSLEngine on
        SSLCertificateFile /etc/apache2/ssl/naw5.crt
        SSLCertificateKeyFile /etc/apache2/ssl/naw5.key

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
    </VirtualHost>
</IfModule>
```

```
  GNU nano 5.4                                  /etc/apache2/sites-ava
<VirtualHost *:80>
    ServerAdmin webmaster@naw5.informatik.hs-furtwangen.de
    ServerName www.naw5.informatik.hs-furtwangen.de
    DocumentRoot /var/www/html

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>

<VirtualHost [::]:80>
    ServerAdmin webmaster@naw5.informatik.hs-furtwangen.de
    ServerName www.naw5.informatik.hs-furtwangen.de
    DocumentRoot /var/www/html

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

(webmaster@localhost) habe ich wider zurück verendert

Schritt 5 – IPv6-Unterstützung konfigurieren

Ich habe in der ports.conf das eingetragen so das auf den ports gelauscht wird

Listen 0.0.0.0:80

Listen [::]:80

<IfModule ssl_module>

    Listen 0.0.0.0:443

    Listen [::]:443

</IfModule>

Schritt 6: Aktivieren von HTTPS und den Konfigurationen

sudo a2enmod ssl

sudo a2ensite default-ssl.conf

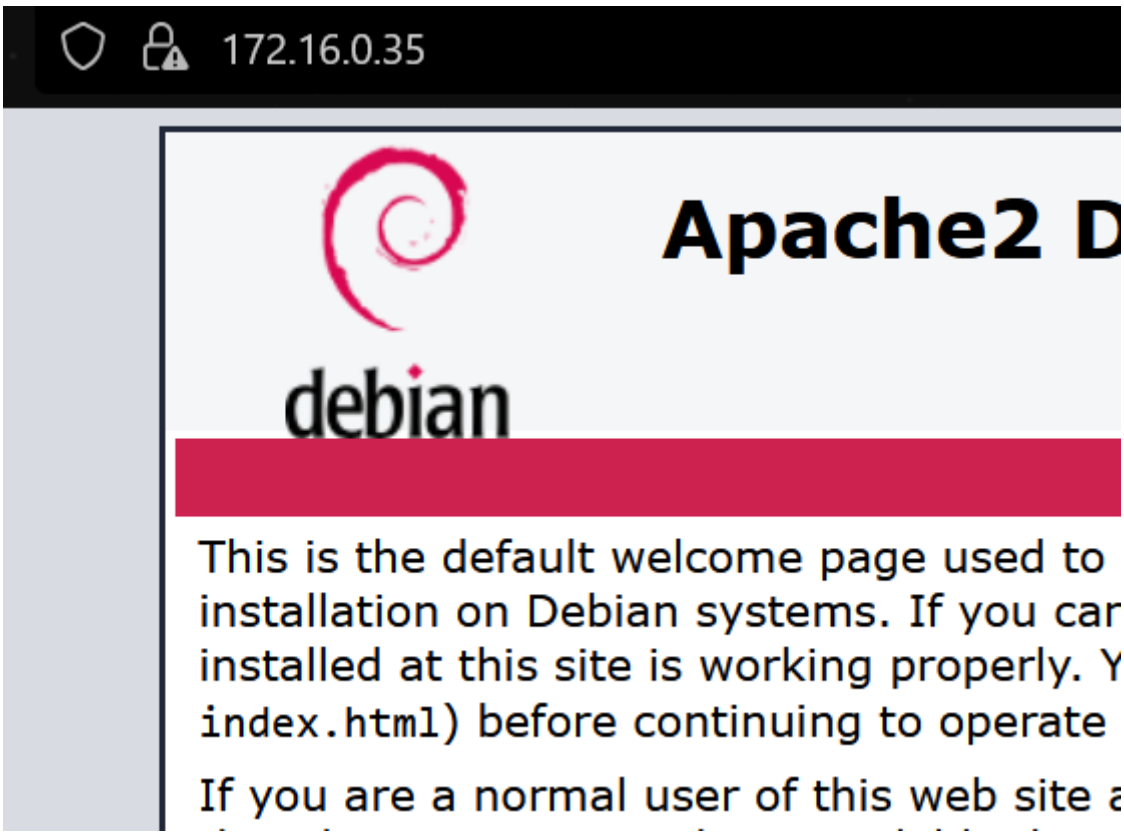
sudo systemctl reload apache2

Schritt 7: DNS-Eintrag / test

```
C:\Users\Kevin_W>nslookup www.naw5.informatik.hs-furtwangen.de
DNS request timed out.
    timeout was 2 seconds.
Server:  UnKnown
Address:  2001:7c0::53:1

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
^C
C:\Users\Kevin_W>nslookup -4 www.naw5.informatik.hs-furtwangen.de
*** Ungültige Option: 4.
DNS request timed out.
    timeout was 2 seconds.
Server:  UnKnown
Address:  2001:7c0::53:1

^C
C:\Users\Kevin_W>nslookup www.naw5.informatik.hs-furtwangen.de 172.16.0.35
Server:  UnKnown
Address:  172.16.0.35

Name:    www.naw5.informatik.hs-furtwangen.de
Addresses:  fe80::cad3:ffff:feb7:e3ec
          172.16.0.35


C:\Users\Kevin_W>
```

```
 EDNS: version: 0, flags:; udp: 1232
 COOKIE: ad6f720aedf68aea010000006840571ddebdface7b5db3e3 (good)
; QUESTION SECTION:
www.naw5.informatik.hs-furtwangen.de. IN A

; ANSWER SECTION:
ww.naw5.informatik.hs-furtwangen.de. 604800 IN A 172.16.150.80

; Query time: 0 msec
; SERVER: ::1#53(::1)
; WHEN: Wed Jun 04 16:24:29 CEST 2025
; MSG SIZE  rcvd: 109

tudent@debian:~$ dig @localhost www.naw5.informatik.hs-furtwangen.de AAAA

 <<>> DiG 9.16.50-Debian <<>> @localhost www.naw5.informatik.hs-furtwangen.de AAAA
 (2 servers found)
; global options: +cmd
; Got answer:
; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 17465
; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

; OPT PSEUDOSECTION:
 EDNS: version: 0, flags:; udp: 1232
 COOKIE: 20cc71b7f93dc61d010000006840573ca4927cb3fd361540 (good)
; QUESTION SECTION:
www.naw5.informatik.hs-furtwangen.de. IN AAAA

; ANSWER SECTION:
ww.naw5.informatik.hs-furtwangen.de. 604800 IN AAAA 2001:7c0:f00:2150::80

; Query time: 0 msec
; SERVER: ::1#53(::1)
; WHEN: Wed Jun 04 16:25:00 CEST 2025
; MSG SIZE  rcvd: 121

tudent@debian:~$
```

Schritt 8: Apache anpassen


172.16.0.35
Apache2 D
debian
This is the default welcome page used to
installation on Debian systems. If you car
installed at this site is working properly. Y
index.html) before continuing to operate
If you are a normal user of this web site a